Requirements Gathering

Stakeholder Analysis

Stakeholder	Needs and Executions
Legal Authorities Police, Emergency Services	High accuracy (low false alarms), real time alerts with precise location data, and video evidence for prosecution.
Schools & Universities	Rapid detection of fighting or vandalism, integration with existing security hardware, and maintaining a safe campus without needing a 24/7 human watching.
Parents	Assurance of child safety, and immediate notification in life-threatening scenarios.
System Administrators	Easy to use dashboard, low latency, and scalable architecture to handle multiple camera feeds simultaneously.

User Stories & Use Cases

User Stories

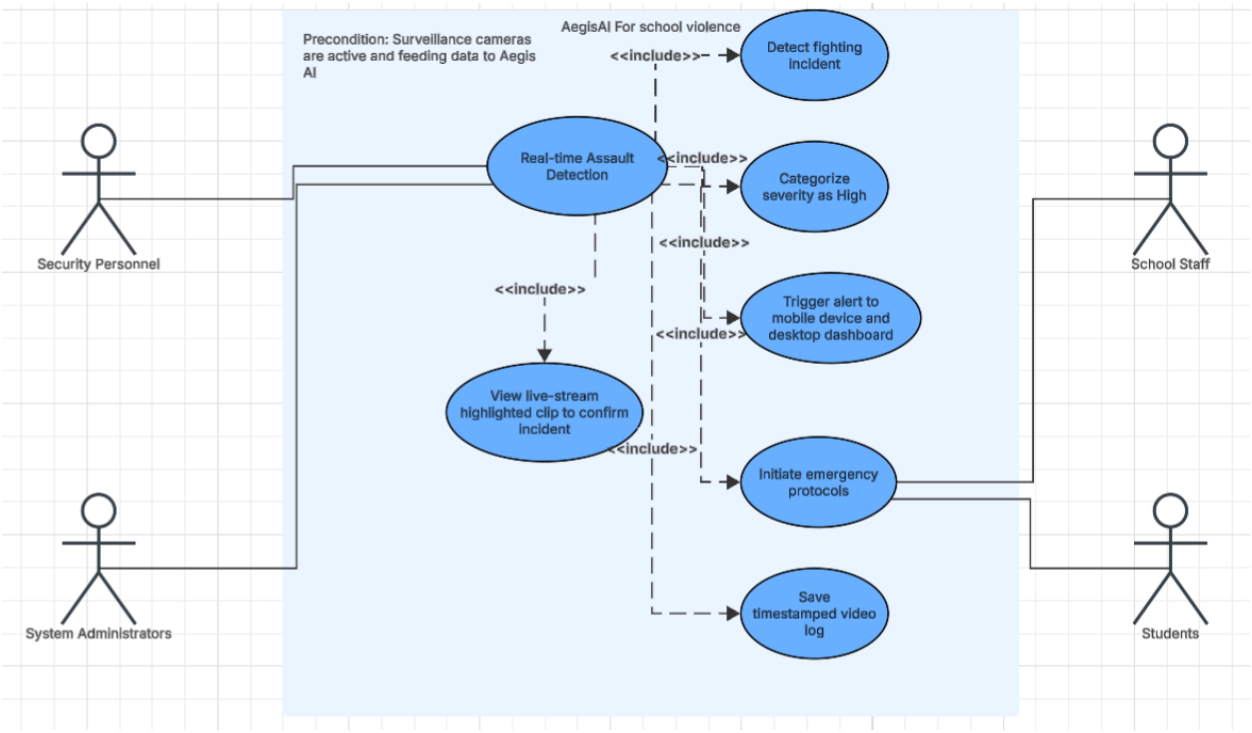
As a Police Officer: I want the system to automatically flag "Shooting" , "Explosion" incidents or any violation with a live video feed, so that I can send the correct tactical unit to the scene without waiting for a call.

As a School Principal: I want to be notified of "Vandalism" or "Fighting" on campus after seconds so that I can prevent property damage and student injury.

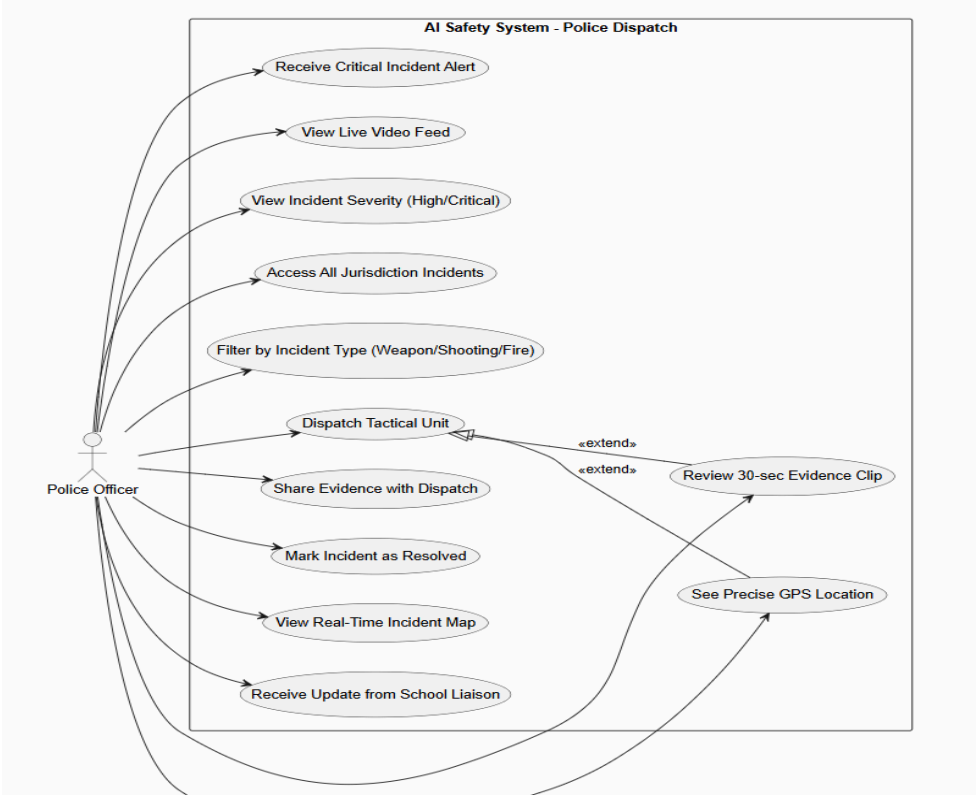
As a Parent: I want to know that the AI is monitoring the surrounding environment for "Abduction" or "Weapon" threats to ensure my child is in a secure environment.

User case

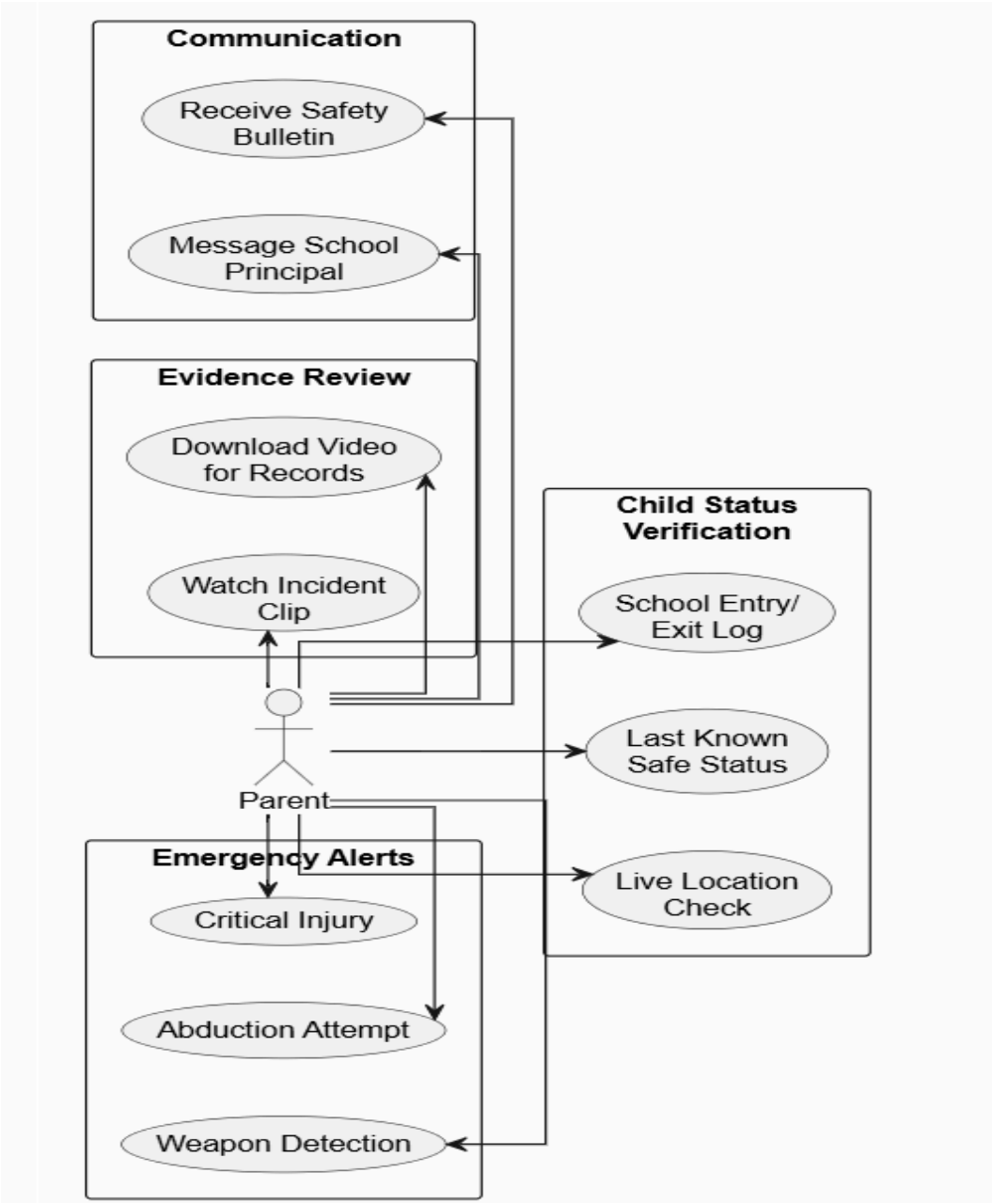
School Violence:



Incident:



Parents:



Functional Requirements FRs

FR-1: Real-time Detection: The system shall detect 13+ specific incidents (Abuse, Arson, Robbery, etc.) in a few seconds of the event occurring.

FR-2: Severity Categorization: The system shall automatically label incidents as Low, Medium, or High priority based on the nature of the detected action.

FR-3: Alert Triggering: The system shall send multi-channel notifications (SMS, Email, Dashboard Push) to designated stakeholders when a threat is identified.

FR-4: Dashboard & Analytics: The system shall provide a web based UI to view live feeds, historical incident logs, and frequency heatmaps.

FR-5: Archiving: The system shall automatically clip and store 30 seconds of video prior to and after a detected incident for legal evidence.

Non-Functional Requirements NFRs

Performance: The system must process video frames at a minimum of 15 FPS (Frames Per Second) to ensure "real time" accuracy.

Security: All video data must be encrypted to prevent unauthorized access or tampering.

Reliability: The platform must maintain uptime, as failure to detect a crime could have life-critical consequences.

Scalability: The architecture must support an increase in camera feeds without a significant drop in detection speed.